use of the relations of type (5), instead of (4), makes it possible to avoid this.

Reader: But how can we use the relations of type (5) without implicitly applying the relations of type (4)?

Author: What is done this.

First, using (4), we find the result of applying the operator $\frac{d}{dx}$ to a sum, product, and ratio of functions, and to composite or inverse functions provided that the result of applying the operator to the initial function (or functions) is known. In other words, the first step is to establish the rules for the differentiation of functions.

Second, using (4), we find out the result of applying $\frac{d}{dx}$ to some basic elementary functions (for instance, $y=x^n$, $y=\sin x$ and $y=\log_a x$).

After these two steps are completed you can practically forget about the relations of type (4). In order to differentiate a function, it is sufficient to express the function via basic elementary functions (the derivatives of which were obtained earlier) and apply the rules for differentiation.

Reader: Does it mean that the relations of type (4) could be put aside after they have been used, first, for compiling a set of differentiation rules and, second, for making a table of derivatives for basic elementary functions?

Author: Yes this is the procedure. Using the differentiation rules and the table of derivatives for some basic elementary functions you are in a position to forget about the relations of type (4) and are free to proceed further by using the "language" of the relations of type (5). A formal course of differential calculus could skip the analysis of limit transition operations, that is, the relations of type (4). It is quite sufficient for a student to learn a set of differentiation rules and a table of derivatives of some functions.